

Example: agent design sheet for brief 4, ticket to work order

What this is: The S24 agent design sheet filled in for use case brief 4, the way a group might write it at the end of Day 4. The plant is the fictional Sabkha Gas Plant (SGP); the pump is P-1201A at MRB, and the May seal job is WO-118305. It shows the level of detail we are after. It is one answer, not the answer, and it leaves real gaps open on purpose. The tools are the four on the reference MCP server in front of the mock ERP. The three caps are estimates, not measurements, because the brief has not been run end to end. Synthetic program material: no real OQ data.

Field	Answer
Author	Example group (brief 4)
System	Brief 4: ticket to work order (agents)
Date	Day 4, 30 September 2026

1. The rung and the four questions

Question (S20)	Answer	So
1. Can I draw the flowchart?	Mostly. Ticket, tag, equipment, history, draft, approval, write. Not the ticket that names no tag.	Rung 4 for the main path
2. Does it need to act, or only to answer?	Act, once: one work order, and never without a planner.	The gate question, section 2
3. If it is wrong, who finds out and when?	The planner, before anything is written. If the planner clicks without reading, the technician, on arrival.	Approval is the ceiling, so the draft must be readable in one screen
4. How does it know it is finished?	The draft is handed to the planner and answered, or a cap bites.	Three caps, section 3

- Our rung, and why.

Rung 4: a workflow with model calls at two steps, extracting the tag and the requested action from the ticket, and writing the draft. We climb to rung 5 only if the tickets that name no tag fail the workflow on the eval. A tag in the text is not always the equipment at fault (some tickets carry tags as distractors, see `plant_tag_distractor` in the label metadata).

- Open gap.

Nobody has counted how many of the 600 tickets name no tag, or how many name the wrong one. That count decides whether rung 5 is ever needed.

2. The tools, and the gate

Tool	Reads or writes	Same system can undo it, same minute?	Cost of a wrong one, while it stands	Control
<code>get_equipment</code>	read	n/a	None. A wrong tag returns a 404 the model can read	policy: tag must match the schema

Tool	Reads or writes	Same system can undo it, same minute?	Cost of a wrong one, while it stands	Control
<code>get_maintenance_history</code>	read	n/a	None, but a missing history makes the draft blind and the planner may not notice	policy
<code>list_work_orders</code>	read	n/a	None. Used to find an existing order for the same ticket	policy
<code>raise_work_order</code>	write	No. The ERP has no edit, cancel or delete	Released to the site crew at once. Priority 1 calls a crew out now	approved: the planner sees the exact body

- Where the gate sits.

In the MCP server: `raise_work_order` does not exist unless writes are switched on, and even then the first call is answered with an approval form for these exact arguments. The retry must carry that approval, once, within 10 minutes, from an approver on the list. In the client: the planner sees the body of the POST, the history that was attached and the tool calls that produced it.

- What the agent may propose.

One work order per source ticket, work type and priority chosen from the ticket and the history. Priority 1 is the planner's to type, not the agent's to propose.

- What happens on a retry.

The ERP answers 409 Conflict for a source ticket that already has an order, so a retry after a timeout cannot raise a second one. The gate uses each approval once.

- Open gap.

The gate proves the server asked about these arguments and got a decision naming an approver. It does not prove a person answered. Who clicked is the host application's login, and we have not chosen it.

3. The three caps

Cap	Our value	How we chose it	What happens when it bites
Steps	12	Estimate: 3 reads, 2 model calls, 2 write rounds is 7; rounded up for retries. Not measured	The run stops, the draft is marked incomplete, the planner sees what was found
Money per run	OPEN	Measure 20 tickets in S27, then double it. Lab 12's workflow arm cost USD 1.60 per 1,000 runs on a shorter job	The client raises before the next call is paid for
Wall clock per run	60 seconds, plus the approval wait	Estimate. The wait is not counted, but an approval expires after 10 minutes	A draft nobody approves in time is logged as not approved

- Open gap.

Ten minutes is the server's limit. A planner who answers after a shift change gets a fresh draft. We have not asked planning whether that is acceptable.

4. The five audit fields, for every write

Field	Can we produce it today?	Where it is written
Who asked	Partly	The ticket's reporter is free text. The caller in the audit line is unverified-client, so the caller id is not evidence
What was proposed	Yes	The first audit line: decision pending, the arguments redacted, the argument hash
Who decided, against which rule	Yes	The second audit line: approver, reason, and the rule that refused if it did
What executed	Yes	Same argument hash on both lines; the result hash and the ERP status 201
What changed on the record	Yes	A read of the ERP for that source ticket: one order, status released, approved_by set

- Open gap.

Two of the five depend on a person's identity that nothing checks yet: who asked, and who clicked. An audit trail cannot be added afterwards to writes that already happened.